

The Network of Modern AI Research: A Quantitative Analysis of Influential Research Communities (2023-2025)

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Interactive Demo: <https://huggingface.co/spaces/hesamation/Daily-Papers-Atlas>

Project Repository: <https://github.com/hesamsheikh/daily-papers-analysis>

1 Introduction

With the rise of large language models (LLMs) and transformer-based architectures, AI research has witnessed substantial growth. New influential papers are published daily by the research community, helping us understand more about these high-scale models. This dynamic research activity asks for a systematic analysis of the current state of AI advancements.

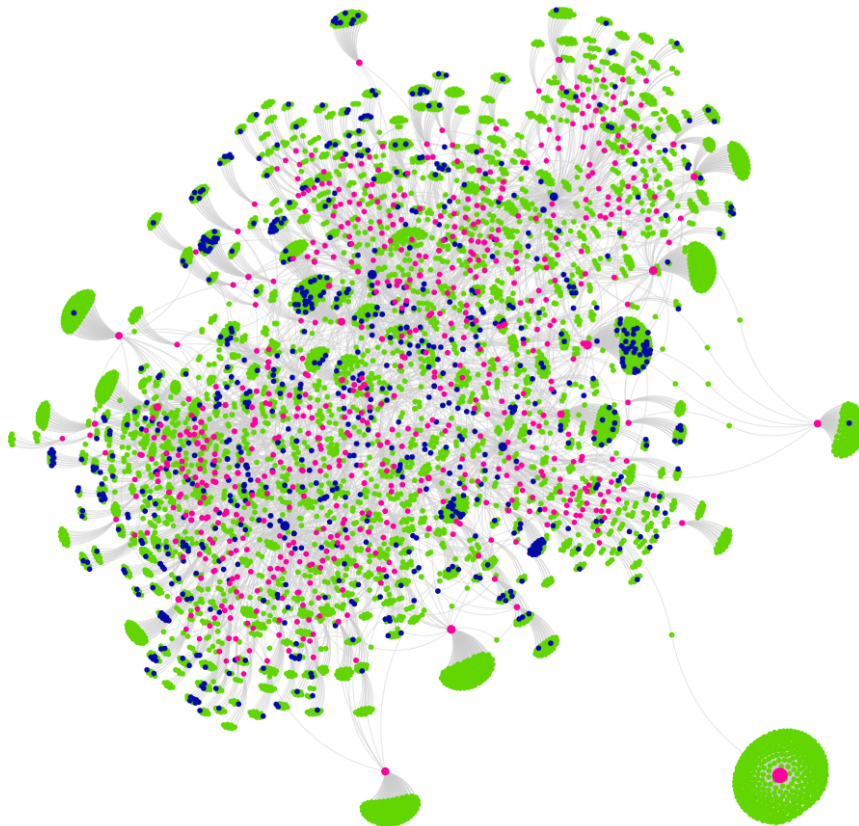


Figure 1: An example network consisting of organizations (blue) and authors (green) with the most papers (4 each), and the top 10 papers (pink) with the most authors and organizations.

This study explores the key research papers featured on the Hugging Face daily papers[1] page. We scraped the paper metadata, including paper details and authors' information, and utilized an LLM to extract the organizations behind these publications (e.g., universities and companies). Using this data, we constructed a graph dataset. Finally, we examined this network to gain valuable insights.

2 Problem and Motivation

The exponential growth in AI research ignited by the advancements of LLMs has created a vast landscape. Due to the availability of LLMs and their novelty, research in this field is widespread; numerous entities contribute to the research by building new models or exploring the existing ones. Additionally, the high computational costs and substantial funding required to train large-scale models have driven increased industry participation, leading to a level of collaboration between academia and industry rarely seen in other scientific fields [2].

This has resulted in a diverse body of work, making it challenging to track key contributors to this research landscape. As a result, understanding the structure of this landscape requires a systematic analysis to identify patterns of collaboration, institutional impact, and the flow of knowledge between various entities. The sheer volume of publications makes it difficult to identify which works are truly influential, which institutions drive major research, and how this research effort is distributed globally. Such a study can help different actors (e.g. countries) to identify how much they truly have an impact in leading or keeping up with these breakthroughs.

To address these challenges, this study aims to construct a structured network of AI research by analyzing papers featured on the Hugging Face Daily Papers page. Unlike studies that examine hundreds of thousands of publications, this research focuses on a selection of the most influential and high-impact papers that inspire a large community. By extracting metadata, institutional affiliations, and research topics, we build a graph dataset that provides a clearer picture of AI research dynamics. This helps us gain a deeper understanding of the field's evolution, helping researchers, industry professionals, and policymakers make more informed decisions about future directions, collaborations, and resource allocation.

3 Datasets

The Daily Papers page of Hugging Face lists the most important papers based on their impact. These papers include a variety of different topics in machine learning, computer vision, novel benchmarks, but mostly large language models. While this collection doesn't definitively cover all of the papers, it does feature the ones that are of high importance in the community. The collection does not include most of the metadata we need to create our dataset, necessitating the need to create our graph dataset and gather the key information we need.

We create this network of AI research in the following steps:

3.1 Listing Papers

We scraped this collection of papers in the time range of 10th May 2023 and 10th Feb 2025. This gives us a list of **7791** paper titles in that time frame of **643** days that are featured on the Daily Papers page. This process is not time-consuming, but rather straightforward.

3.2 Extracting Paper Metadata

With the list of influential papers available, our next objective was to extract the metadata of the papers. We utilized the arxiv Python library to access the arXiv API [3]. Then we used this API to search for papers given their titles. ArXiv would then return the paper metadata, which includes information such as the date of publishing, title, authors, summary, categories, the abstract, and the PDF link of the paper. This search process took 2 hours and 25 minutes in total and would return **4195** paper objects, as the duplicate papers were removed and some titles did not yield results from the arxiv API. The collection of these papers is saved in a **32MB** JSON file that is accessible in the project repository.

In this step, we extract the authors and the subject category of each article from the metadata.

3.3 Extracting Publishing Organizations

The publishing organization of a paper is the universities, research centers, companies, or similar entities with which the authors or the paper are affiliated. As the arxiv API did not include any details as to which organizations contributed to the research paper, extracting this turned out to be our next challenge. We particularly analyzed two methods to extract this information:

1. Using the scholarly [4] Python library to search author names on Google Scholar[5] and find the organization they are affiliated with. However, due to the anti-crawling policies of Google Scholar, the search is intentionally time-consuming and thus impractical.
2. Searching for author information through ORCID public API [6]. This turned out ineffective as for many author names, multiple individuals with the same name would be returned.

Given these limitations, we took extra measures to extract the organizations' information without accessing public APIs, but by extracting it using LLMs. This is time-consuming and computationally expensive, but it is the most reliable method for two reasons: the publicly available APIs may not return the most recent organization affiliations of a person, but the organizations mentioned in the paper are most definitely affiliated with the paper. Additionally, an individual may have multiple organizational affiliations, and resolving which organization is affiliated with a given paper cannot be easily resolved.

Thus, we extract organization information from papers in two steps:

3.3.1 Downloading Papers

To parse the papers with LLM, we used the PDF link from the paper metadata to download the full papers, resulting in 30GB of saved files. This process took several hours to complete.

3.3.2 LLM Extraction

We used Ollama[7] to serve LLMs locally. Ollama provides tools to download, run, and host various large language models with minimum effort. A local device with an Nvidia GeForce RTX 2060 was used to serve the LLM. For the limited computational budget and relative simplicity of the task at hand, we tested various lightweight LLMs and decided to process the PDFs using Qwen2.5-3B[8].

To limit the number of input tokens, we only extracted the text of each paper that would appear before the *Abstract* section. Processing all papers with the LLM took approximately 4 hours to complete, and the model, in most cases, was able to return the correct organization names, even though it sometimes detected false positives (such as the address of an organization).

Prompt

You will be provided with a segment of text extracted from the beginning of a research paper (before the abstract). Your task is to extract a list of unique organization names mentioned in the text. Follow these instructions exactly:

1. **Extract Only Organizations:** Identify and extract only the names of organizations. Do not include:
 - Addresses (e.g., lines that contain location details like "Bethesda, MD, USA")
 - Email addresses
 - Author names or other non-organization text
2. **Hints for Organization Names:** Organization names usually include keywords such as University, Institute, College, Laboratory, Center, Corporation, Inc., Ltd., etc.
3. **Formatting:**
 - List each organization on its own line.
 - Do not include any additional text, commentary, or punctuation.
 - If an organization name spans multiple lines, combine them into one single name (without the extra line breaks).
4. **Avoid Duplicates:** If an organization is mentioned more than once, list it only once.
5. **No Organizations Found:** If no organization names are identified, output exactly:

No organizations found

Now, process the following text:

3.4 Graph Dataset Format

We use the CSV format to save the graph datasets. Different entities and relationships between them are saved in separate files: paper, author, category, organization, paper-author relationship, paper-category relationship, and paper-organization relationship. This means we can create three graphs (using the relationships) and one whole graph that includes all entities and their relations, which will be referred to as **the core network** in this report. We can also infer additional graphs using these files (e.g., from the paper-author graph, we can create an author-author graph in which co-authors are connected).

Type	Entity	Number
Node	Paper	4194
Node	Author	18523
Node	Category	224
Node	Organization	3669
Edge	Paper-Author	28675
Edge	Paper-Category	8601
Edge	Paper-Organization	9783
Edge	Author-Author	241405
Edge	Organization-Organization	14572

Table 1: The number of nodes and edges in the constructed graph datasets.

4 Validity and Reliability

The validity of the dataset used in this study is ensured by sourcing it from the Daily Papers page, a widely recognized platform that highlights high-impact AI research papers. This dataset prioritizes influential publications based on their reception within the research community. However, since it does not comprehensively cover all AI-related publications, selection bias may be present. While this dataset provides a faithful representation of AI research collaborations, certain limitations remain, such as missing metadata and evolving author affiliations, which may affect completeness.

The reliability of this study is addressed through a systematic and reproducible methodology for processing and analyzing AI research collaboration networks. The dataset was processed using *pandas* for data handling and *networkx* for network construction. The graph construction process followed a structured pipeline, such as duplicate removal. Network analyses, including degree centrality, modularity, and community detection, were conducted using deterministic algorithms where feasible. For stochastic methods, random seeds were explicitly set to ensure repeatability. While minor variations may arise due to API updates, the overall methodology guarantees consistency and reliability.

Due to the probabilistic nature of LLMs, the extraction of organizations from the papers may yield slightly different results when running the process again or using a different large language model. Leveraging bigger models, potentially with reasoning capabilities, ensures a more accurate result, but also higher computational demands. However, our qualitative survey of the extracted organizations shows that the overall process is reliable and trustworthy.

5 Measures and Results

By constructing and analyzing three distinct graphs, The Core Network, Co-authorship Network, and Co-organization Network, we aim to uncover the intricate relationships and patterns that define their collaboration.

The Core Network integrates authors, organizations, and papers, providing a holistic view

of the research ecosystem and highlighting key players and interactions. The **Co-authorship Network** focuses on the connections between authors and their publications, revealing patterns of co-authorship and individual contributions. Finally, the **Co-organization Network** examines the ties between organizations and papers.

Overall, these networks offer a multi-dimensional perspective on the research community, enabling us to identify central authors, collaborative organizations, and quantitative analysis of the published papers. Through this analysis, we seek to better understand how knowledge is created, shared, and propagated within this network.

5.1 The Core Network

In this section, we examine the Core Network, a structured graph that connects papers, authors, and organizations. Figure 2 presents a visualization of this network, showing the relationships between these entities and highlighting the most influential nodes.

We analyze the structural properties of the network by analyzing its degree distribution, robustness, and key centrality measures. A highly skewed degree distribution suggests the presence of influential figures, while robustness analysis evaluates the resilience of the network to node removal. Furthermore, centrality measures such as degree centrality and betweenness centrality help identify key contributors within the AI research landscape.

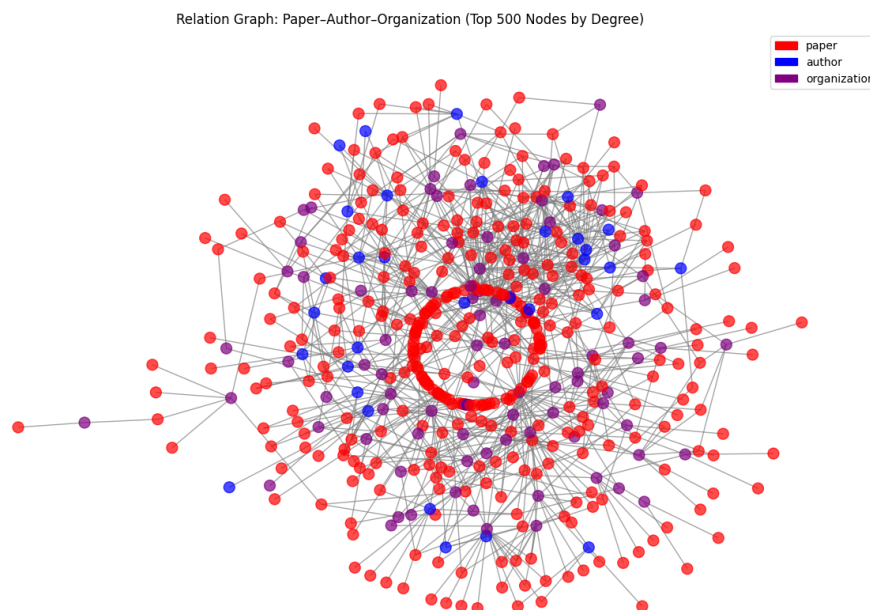


Figure 2: Paper-Author-Organization Collaboration Network (Top 500 Nodes by Degree)

5.1.1 Degree Distribution Analysis

The degree distribution of the network provides insight into its structural properties and connectivity patterns. As shown in Figure 3, the distribution exhibits a highly skewed nature, where the majority of nodes have a low degree, while a small fraction maintains significantly higher connectivity. This pattern indicates a power-law distribution. The result gave an exponent of

around $\alpha \approx 3.84$. This confirms that the network has a heavy-tailed structure. However, because the exponent is higher than the typical range for scale-free networks ($2 < \alpha < 3$), the network does not follow the usual scale-free pattern.

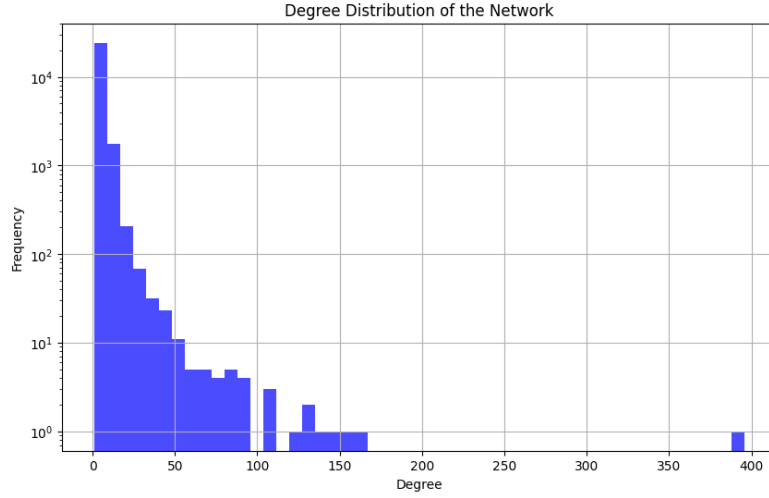


Figure 3: Degree Distribution of the Core Network

5.1.2 Log-Log Degree Distribution

The plot of the degree distribution on a log-log scale, shown in Figure 4, reveals that the network has a heavy-tailed structure, where most nodes have few connections and a smaller number have significantly more. The middle part of the curve appears roughly linear, suggesting that the network partially follows a power-law distribution. However, the curve bends downward toward the end, indicating a faster drop in the tail. This observation is supported by the estimated power-law exponent, $\alpha \approx 3.84$, which is higher than the typical range for scale-free networks.

When comparing the power-law model to other types of distribution, it fits better than the exponential and stretched exponential models. However, there was no significant difference between the power-law, log-normal, and truncated power-law fits. This means the network is heavy-tailed, but it may follow a log-normal or truncated power-law distribution instead. In other words, while some papers, authors, and organizations are more connected than others, their influence is less extreme than in typical scale-free networks.

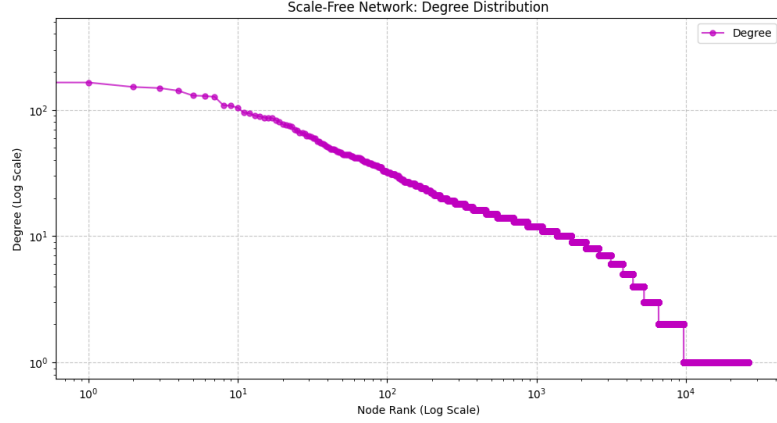


Figure 4: Log-Log Degree Distribution of the Core Network

5.1.3 Network Robustness Analysis

To assess the robustness of the network, we simulated node removal using three strategies: random removal, highest-degree nodes, and highest-betweenness nodes. For each method, up to 40% of the nodes were removed in steps, and the size of the largest connected component was recorded.

As shown in Figure 5, the network remains largely connected under random failures, indicating strong resilience. However, when central nodes (by degree or betweenness) are removed, connectivity declines sharply, with the network nearing collapse after the removal of just 10% of the nodes. This shows that the network is robust to random failures but highly vulnerable to targeted attacks, a typical pattern in heavy-tailed networks.

While this may resemble the behavior of small-world networks, further analysis confirmed that the network does not exhibit small-world properties. Specifically, it has a clustering coefficient of 0 and an Omega value of $\omega = 1.3644$, indicating that the network is not small-world, but instead has a heavy-tailed structure with moderate hub dominance.

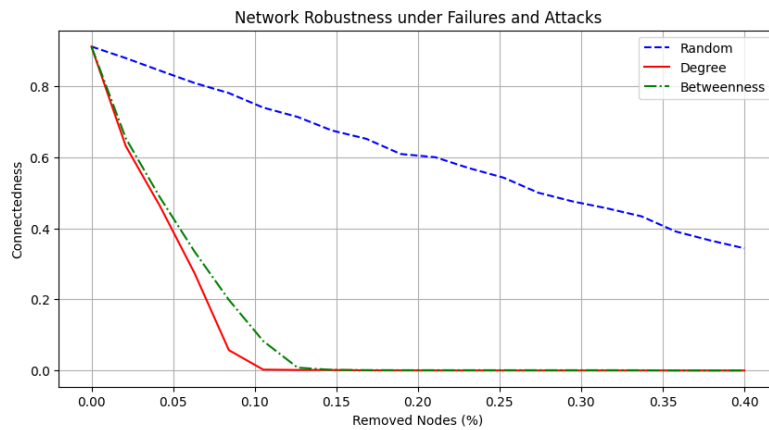


Figure 5: Effect of Node Removal on the Largest Connected Component

5.1.4 Centrality Analysis of Key Nodes

In exploring different aspects of the network, we examined four centrality measures—degree, betweenness, and closeness—across three node types: papers, organizations, and authors.

5.1.4.1 Degree Centrality

Degree centrality quantifies a paper's structural connectivity based on its number of direct connections within the network. Figure 6 highlights the most connected research papers, with *"HyperCLOVA X Technical Report"* and *"Introducing v0.5 of the AI Safety Benchmark from MLCommons"* showing the highest connectivity.

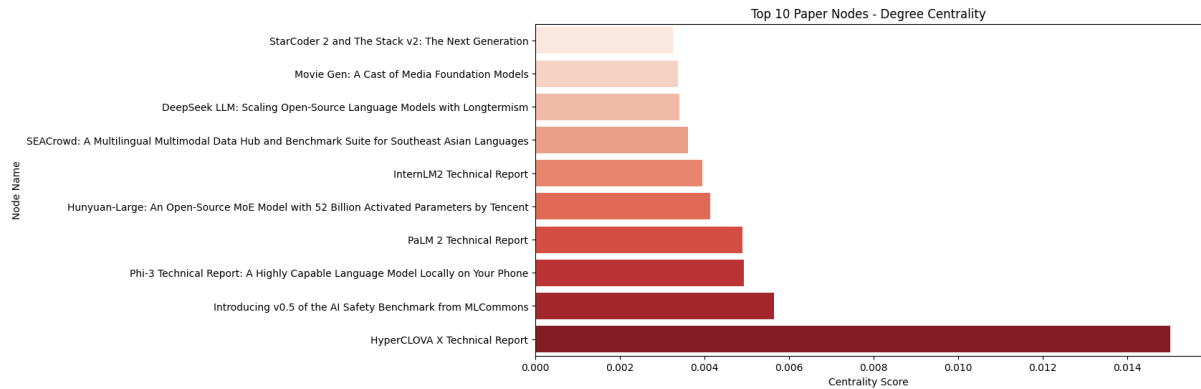


Figure 6: Top 10 papers by Degree Centrality in the Network

5.1.4.2 Betweenness Centrality

Betweenness centrality quantifies the role of a paper in facilitating knowledge flow by acting as a bridge between different research actors. Figure 7 highlights the papers with the highest betweenness scores. Papers such as *"Introducing v0.5 of the AI Safety Benchmark from MLCommons"* and *"HyperCLOVA X Technical Report"* exhibit the highest centrality. The removal of these papers would significantly impact the connectivity of the network.

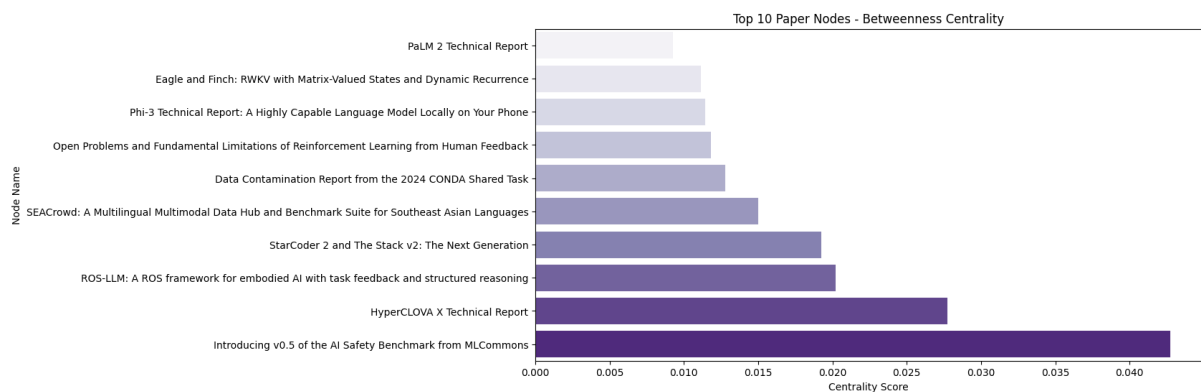


Figure 7: Top 10 Papers by Betweenness Centrality in the Network

5.1.4.3 Closeness Centrality

Closeness centrality measures which papers are most centrally connected in our network. Figure 8 highlights key papers with high closeness centrality. Papers such as *"Introducing v0.5 of the AI Safety Benchmark from MLCommons"*, *"ROS-LLM"*, and *"GMAI-VL-5.5M"* exhibit the highest scores, and show well-connected papers that reach others quickly.

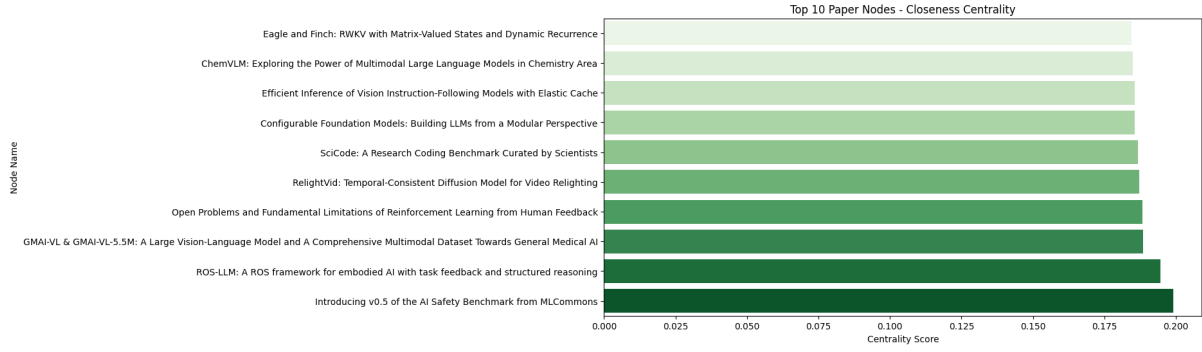


Figure 8: Top 10 papers by Closeness Centrality in the Network

5.2 Co-authorship Network

In this section, we analyze the co-authorship network (Figure 9). It is a weighted network derived from paper-author relationships. Each node in this network represents an author, and edges indicate that two authors co-authored the paper(s), with the weight showing the number of papers they have collaborated. This network reveals how authors collaborate, hinting at the presence of a few highly connected hub authors and multiple specialized communities.

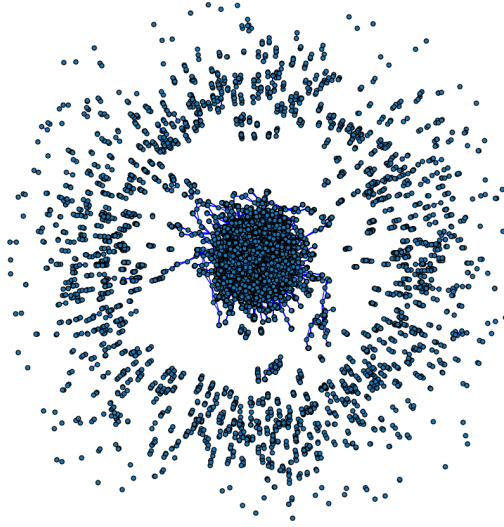


Figure 9: An overall figure of the co-authorship network.

5.2.1 Centrality Measures

To reveal interesting information from this network, we calculate Weighted Degree and Betweenness Centrality. In Figure 10, the Weighted Degree measures show each author's total co-authorship volume (summing all collaboration links). For example, Yu Qiao appears to have the highest total co-authorship count among the authors. It is important to note that weighted degree centrality measures how many co-authors a researcher has worked with, showing who collaborates most frequently. But a high number of collaborations does not always mean greater influence; some may come from routine partnerships or large team projects, like industry reports. The Betweenness Centrality indicates how frequently an author serves as a bridge on the shortest paths between other authors.

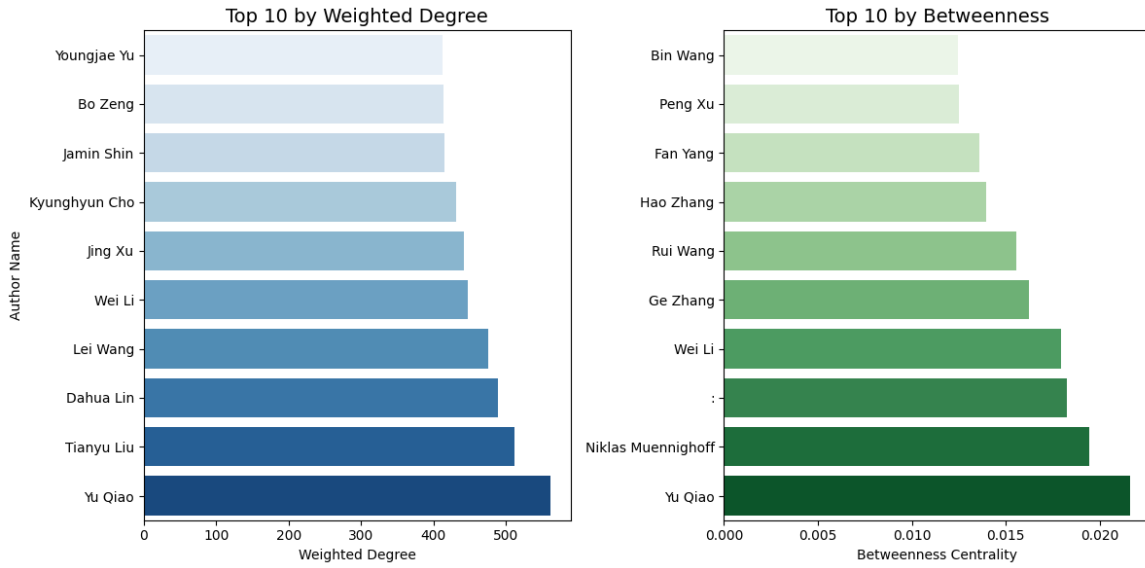


Figure 10: Top 10 authors in a co-authorship network ranked by weighted degree and betweenness centrality.

5.2.2 Modularity

Modularity measures how a network is divided into communities. This network has a strong community structure, with a modularity score of 0.8143. This means authors tend to collaborate more within their groups rather than with others outside their group. Additionally, the network is made up of tightly connected clusters with only a few connections between them.

5.2.3 Community

Our analysis of the co-authorship network uncovered 53 unique communities, by using the Louvain method. The largest of these includes 1.174 authors, making it a major hub for collaboration, likely representing a well-connected research group. This community is made up of 552 organizations, such as *New York University*, *J.P. Morgan AI Research*, *Microsoft*, *Peking University*, and *Georgia Institute of Technology*. On the other hand, the network has many smaller communities (e.g., 5 authors) which might reflect niche research areas, new collaborations, or specialized fields with fewer participants.

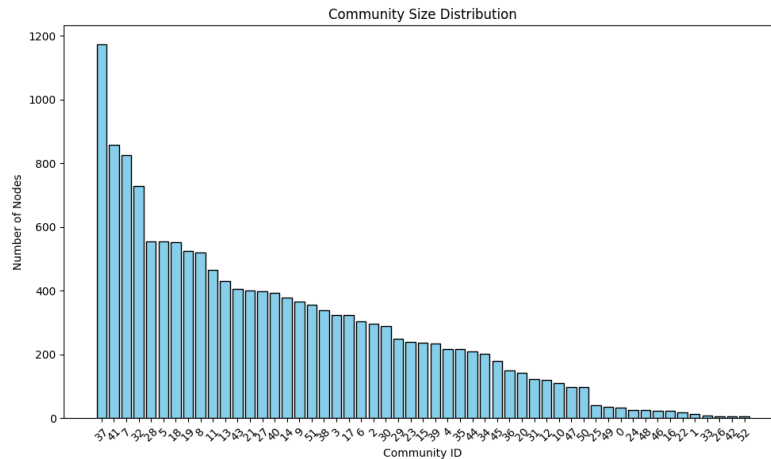


Figure 11: Communities in Co-authorship Network

5.2.4 Global Clustering Coefficient (Transitivity)

The Global Clustering Coefficient is a measure of how tightly connected the nodes in a network are. It quantifies the tendency of nodes to form triangles (groups of three nodes that are all connected). The global clustering coefficient for this network is 0.941, which is very close to the maximum value of 1. This indicates that this network has a highly clustered structure, meaning that authors in the network tend to form tightly connected groups.

5.2.5 Scale-free

The degree distribution of the co-authorship network follows a power-law distribution with an alpha value of 1.87, indicating a scale-free structure, which means that it consists of a few authors that are highly connected, acting as central hubs, while most authors have fewer collaborations. This suggests that the network is shaped by a small number of key players who collaborate extensively, while the majority of authors work with fewer partners.

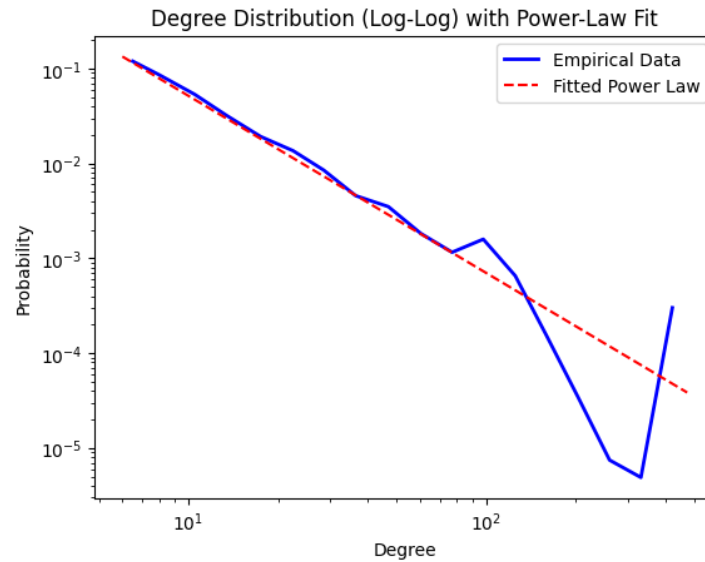


Figure 12: Power-law in Co-authorship Network

5.2.6 Clique

A clique is a group of nodes where every pair of nodes is directly connected, and the maximal cliques are those that cannot be extended by adding any other node from the graph without losing the clique property. The size of the largest clique is 396, which reveals the paper with the most authors, since authors of a paper naturally form a clique.

Subgraph of the Largest Clique

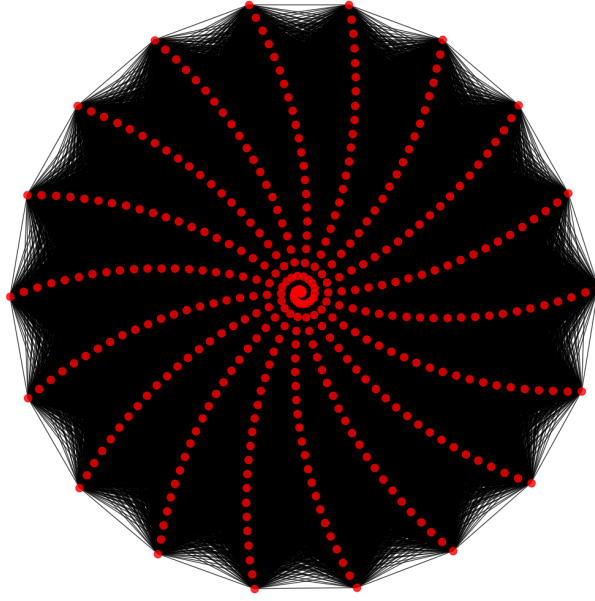


Figure 13: Largest Clique in Co-authorship Network

5.3 Co-organization Network

In this section, we analyze the Co-organization network. It is a weighted network derived from paper-organization relationships. Each node in this network represents an organization, and edges indicate that two organizations have collaborated, with the weight indicating the number of collaborations.

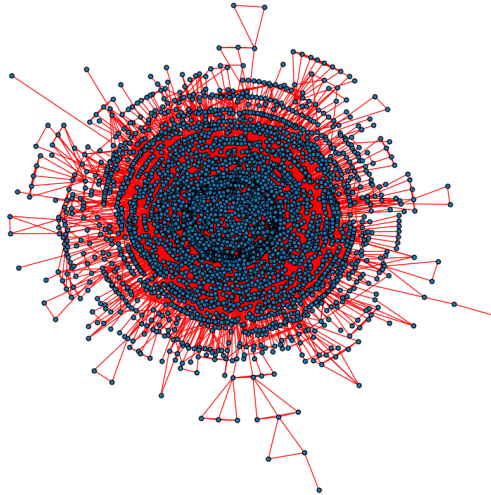


Figure 14: Co-organization network consists of 3669 nodes and 14572 edges

5.3.1 Centrality Measures

Similar to the co-author network, we calculate Weighted Degree and Betweenness Centrality. In Figure 15, the Weighted Degree metric represents the total collaboration volume of each organization; organizations with high weighted degrees are highly active in collaborative research, but it does not indicate whether collaborations are impactful. The Betweenness Centrality metric measures how often an organization acts as a bridge on the shortest paths between other organizations. Organizations with high betweenness centrality play a critical role in connecting different parts of the network.

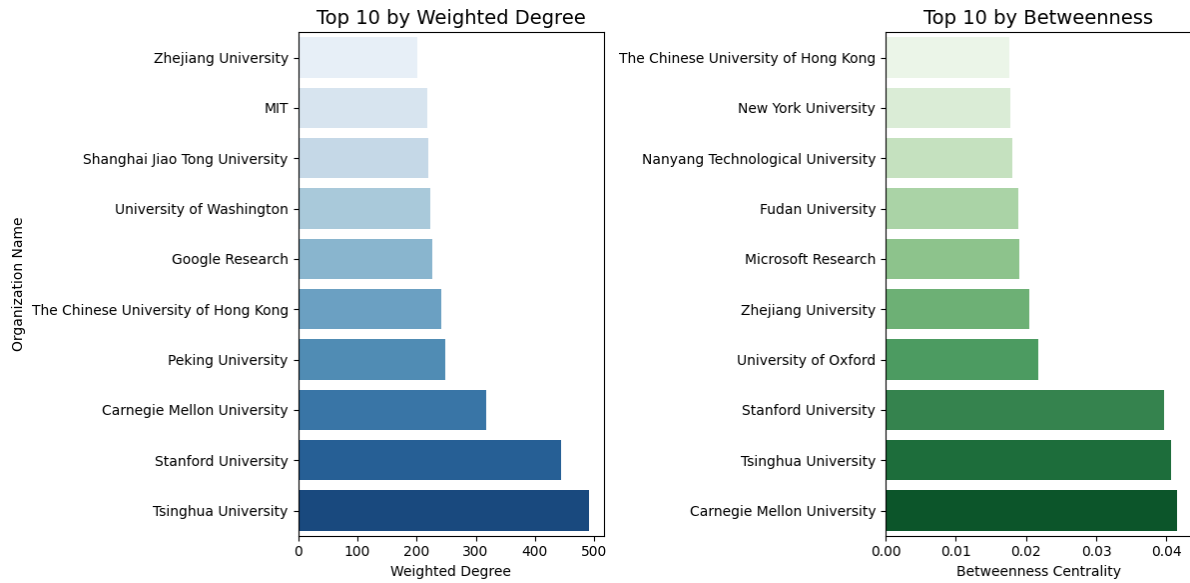


Figure 15: Top 10 organizations in a co-organization network ranked by weighted degree and betweenness centrality.

5.3.2 Modularity

Modularity measures how a network is divided into communities. This network has a strong community structure, with a modularity score of 0.5948. Therefore, the network has distinct and well-defined clusters where organizations within the same cluster collaborate more frequently with each other than with organizations outside their cluster.

5.3.3 Community

Our analysis of the co-organization network uncovered 34 unique communities by using the Louvain method. The community size varies significantly, ranging from 3 to 620 nodes. By exploring the biggest community, we conclude that *Tsinghua University* consists of 165 papers, and the organization in the smallest community is *DP Technology*, with 2 papers.

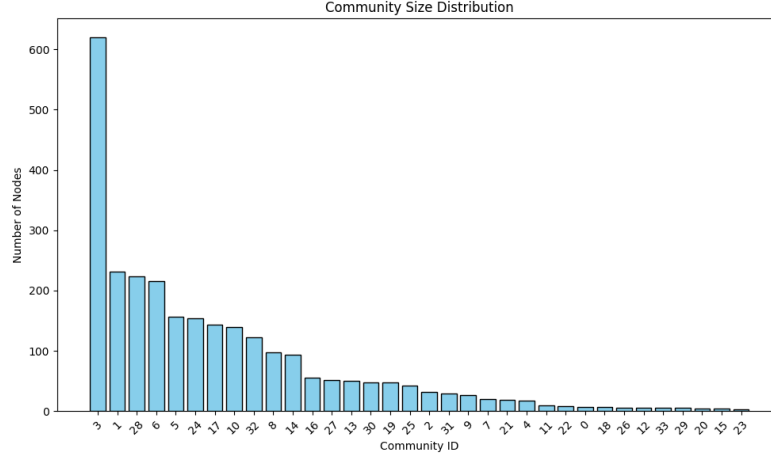


Figure 16: Communities in Co-organization Network

5.3.4 Global Clustering Coefficient (Transitivity)

Global Clustering Coefficient is a tendency for organizations to form tightly-knit collaborative triangles (where two collaborators of an organization are also likely to collaborate). The global clustering coefficient for this network is 0.287, which shows that there is a tendency to form tightly-knit collaborative triangles, but the network is not highly clustered.

5.3.5 Scale-free

The analysis of the degree distribution using the power-law reveals that the network exhibits a scale-free property, with an alpha value of approximately 2.08. This alpha value indicates the presence of a few highly connected nodes and many nodes with low degrees, which is typical in scale-free networks.

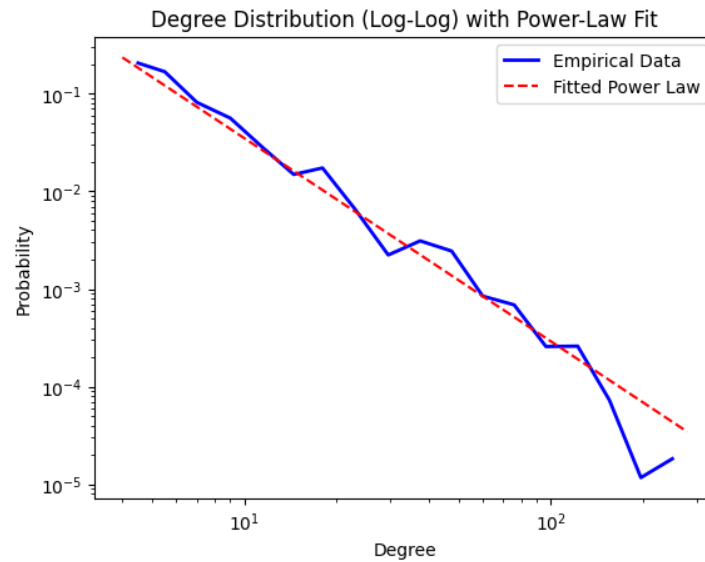


Figure 17: Power-law in Co-organization Network

5.3.6 Clique

The largest clique in this network includes 49 nodes, relating to the paper *Introducing v0.5 of the AI Safety Benchmark from MLCommons*, which has the largest number of organizations

involved in the research process.

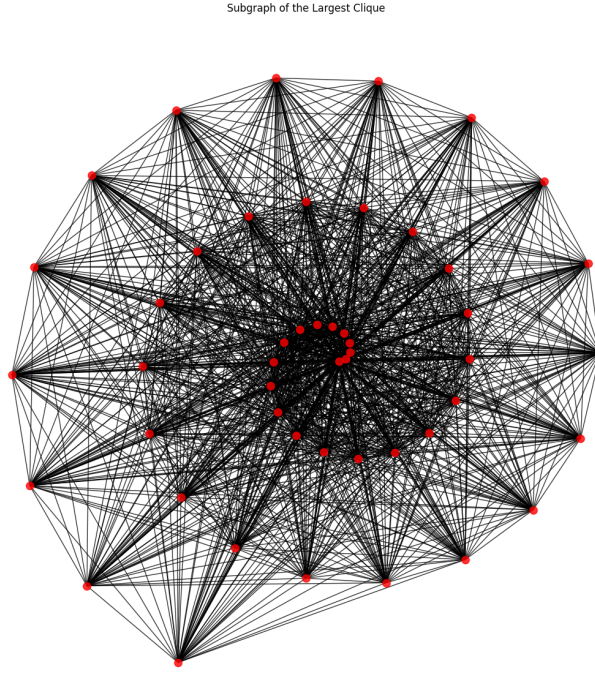


Figure 18: Largest Clique in Co-organization Network

6 Conclusion

This study presents a structured network analysis of modern AI research, focusing on the important papers, organizations, and collaborations within the Hugging Face Daily Papers. By constructing and examining three distinct networks—the Core Network, Co-authorship Network, and Co-organization Network, we provide a comprehensive view of the AI research landscape.

Our results show that the core network follows a heavy-tailed structure, where most papers, authors, and organizations have only a few connections, while a smaller number are highly connected and play a key role in linking different parts of the network. However, the network does not fully conform to a typical scale-free pattern, as the influence of these central nodes is present but not extreme. In contrast, the Co-authorship Network and the Co-organization Network follow a scale-free structure.

Centrality analysis identifies structurally prominent papers, including "*HyperCLOVA X Technical Report*" and "*Introducing v0.5 of the AI Safety Benchmark from MLCommons*". These papers act as reference points in AI research, influencing many different fields. Our analysis of co-authorship and co-organization networks also shows that leading organizations like "*Tsinghua University*" and "*Google Research*" are at the center of AI research collaborations. We distinguish this structural prominence from scientific influence, which requires content analysis.

Our robustness analysis of the core network shows that it stays resilient to random failures

but breaks apart quickly when central or highly connected nodes are removed. Although the network does not follow a typical scale-free pattern, it is still structurally dependent on a small group of influential nodes. This makes the network vulnerable to targeted disruptions.

Community detection shows that AI research is highly modular, meaning that it is divided into specialized clusters. The largest research community includes 1,174 authors and involves organizations such as "New York University", "J.P. Morgan AI Research", "Microsoft", and "Peking University". This suggests that AI research is divided into many focused areas of study.

Our study answers the research motivation by providing a clear and structured way to track AI research trends. By identifying key contributors and collaboration patterns, we offer useful insights for researchers, industry professionals, and policymakers. These insights help in choosing future research directions, finding collaboration opportunities, and allocating resources wisely.

7 Critique

The study successfully analyzes the research landscape of the most influential AI papers of the previous years. Our graph dataset, which consists of papers, authors, organization, and their various relationships, can be a key feature for other researchers to survey this network from different angles. Additionally, our method of creating the dataset is also useful for those who are interested in analyzing AI papers.

While our effort to create the dataset included overcoming many challenges and limitations imposed by the open APIs, we succeeded in assembling the dataset through multiple steps. One of these steps, extracting the organizations of a paper using LLMs, has the potential to improve by ensuring the LLM is identifying the organizations with high accuracy. It can be a further improvement to know what the quantitative accuracy of the LLM is in identifying those organizations, however, this requires labeled datasets of papers and their organizations and remains an area for future improvements. Further, identifying which organizations belong to academia and which to the industry can be highly insightful to understand the distribution of model AI research among various entities.

The analysis successfully answers some key questions regarding the current state of AI research, the role of industry and academia in pushing it forward, the concentration or distribution of research among actors, identifying key players, and the quantitative relationships between various entities. However, we believe the dataset has much more room for exploration, defining further questions, and finding their answers.

Overall, this study lays strong foundations for further studies that aim to analyze AI research papers in different contexts, through different metrics, or to answer a new set of questions. This work provides a transparent analysis of various key fundamental questions regarding the current state of AI research, encouraging more researchers to explore this idea or define new ones.

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